Essential in ETABS (40 Hours)

S.No	Session	Topics Covered	Type	Duration (Hrs)
1	Session 1	Overview of Structural Analysis and Design Introduction Workflow of ETABS ETABS User Interface Importing Various Files into ETABS Exercises included for Practice	Theory/Practical	2.0 Hrs
2	Session 2	Excercise included for practice	Practical	2.0 Hrs
3	Session 3	**Creating New Project**: New Model Quick Template, Edit Stories and Grid System, Add Grid at Selected Joints, Grid Options **Material Properties**: Defining Materials, Material Property Data Exercises included for Practice	Theory/Practical	2.0 Hrs
4	Session 4	Excercise included for practice	Practical	2.0 Hrs
5	Session 5	**Section Properties:** Frame Sections, Slab Section, Deck Sections, Wall Sections & Reinforcing Bar Sizes **Modeling:** Draw Grids, Draw Dimension Lines, Draw Joint Objects, Draw Beam/Column/Brace Objects, DrawFloor/Wall Objects, Quick Draw Tools, Draw Wall Openings Modeling: Draw Reference Points, Draw Reference Planes, Draw Wall Stacks Exercises included for Practice	Theory/Practical	2.0 Hrs
6	Session 6	Excercise included for practice	Practical	2.0 Hrs
7	Session 7	**Object Selection Tools:** Select Tool, Deselect Too **Replicate**: Linear, Radial, Mirror, Story **Extrude:** Joint to Frame, Frame to Shell Merge Joints, Align Joints/Frames/Edges, Move Joints /Frames/Edges **Edit Frames:** Divide Frames, Merge Frames, Reverse Frame Connectivity, Modify/Show Frame Type	Theory/Practical	2.0 Hrs
8	Session 8	Excercise included for practice	Practical	**2.0 Hrs**
9	Session 9	**Edit Shells:** Divide Shells, Merge Shells, Expand/Shrink Shells, Split Shell Edge, Remove Joints from Shells,Chamfer Slab Corners, Reverse Wall Local 3 Axis, Modify/show Wall Curve Type Assigning Section Properties **Frame Property Specification:** Section Property, Property Modifier, Releases/Partial Fixity, End Length Offsets,Local Axes & Output Station Support Specification Exercises included for Practice	Theory/Practical	2.0 Hrs
10	Session 10	Excercise included for practice	Practical	2.0 Hrs
11	Session 11	**Load Pattern :** Define Load Pattern, Wind Load Pattern, Shell Uniform Load Sets **Loads:** Joint Loads, Frame Loads, Shell Loads Load Cases, Load Combinations Manual Calculations (Dead Load & Live Load) Set Load Cases to Run, Check Model, Analysis **Display Result:** Deformed Shapes, BMD, SFD, Shell Stresses, Tabular Reports, Support Spring Reactions, DisplayUndeformed Shape, Joint Load Assigns, Frame Load Assigns, Shell Load Assigns	Theory/Practical	2.0 Hrs
12	Session 12	Excercise included for practice	Practical	2.0 Hrs
13	Session 13	**Concrete Frame Design:** Introduction, Design Procedure, View/Revise Preferences, View/Revise Overwrites, Select Design Groups, Select Design Combinations, Start Design Check, Interactive Design, Display Design Info, Change Design Section, Verify Analysis Vs Design Section, Verify All Members Passed Group Definitions **Shear-Wall Design:** Introduction, Design Procedure	Theory/Practical	2.0 Hrs
14	Session 14	Excercise included for practice	Practical	2.0 Hrs
15	Session 15	**Steel Frame Design:** Introduction, Design Procedure, View/Revise Preferences, View/Revise Overwrites, SelectDesign Combinations, Lateral Bracing, Set Lateral Displacement Targets, Start Design/Check and InteractiveDesign, Change Design Section & Make Auto Select Selection Null	Theory/Demo	2.0 Hrs
16	Session 16	Excercise included for practice	Practical	2.0 Hrs
17	Session 17	**Steel Connection Design:** Introduction, Design Procedure **Steel Joist Design:** Introduction, Design Procedure, View/Revise Preference, View/Revise Overwrites, SelectDesign Groups, Select Design Combinations, Start Design/Check, Interactive Design, Change Design Section, Make Auto Select Section Null & Verify Analysis Vs Design Section	Theory/Practical	2.0 Hrs

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18	Session 18	Excercise included for practice	Practical	2.0 Hrs
19	Session 19	**Detailing:** Detailing Preferences, Concrete Component Preferences, Steel Component Preferences, Rebar Selection Rules, Start Detailing, Show Detailing, Clear Detailing, Export Drawing	Theory/Practical	2.0 Hrs
		Custom Section Designer: Introduction, Section Designer Workspace		
		Export		
		Report Creation		
20	Session 20	Excercise included for practice	Practical	2.0 Hrs